Year 2 Pure Mathematics

Chapter 5

Radians

Practice Booklet

Contents

5.1 Radian measure ..... book p. 35
5.2 Arc length ..... book p. 36
5.3 Areas of sectors and segments ..... book p. 37
5.4 Solving trigonometric equations ..... book p. 39
5.5 Small angle approximations ..... book p. 40
Problem solving: Set A ..... book p. 41
Problem solving: Set B ..... book p. 42
Selected exam question bank questions .....
Answers .....

5.1 Radian measure

Practice Book, printed page 35.

1. Convert these angles from radians to degrees, giving your answers to 1 decimal place, where appropriate.

a. $\frac{5\pi}{6}$

b. $\frac{\pi}{8}$

c. 4π

d. 0.34 rad

e. $\sqrt{2}$ rad

f. 1.9 rad

Hint

To convert radians to degrees multiply by $\frac{180}{\pi}$

2.

a. Convert the following angles to radians, giving your answers as multiples of π.

i. 18°

ii. 32.5°

iii. 320°

Hint

To convert degrees to radians multiply by $\frac{\pi}{180}$

b. Convert the following angles to radians, giving your answers to 3 significant figures.

i. 16°

ii. 124°

iii. 260°

3. Sketch the following graphs, marking any points where the graphs intersect with the coordinate axes.

a. $y = \tan 2x$ for $0 \leq x \leq \pi$

b. $y = \cos(x + \pi)$ for $0 \leq x \leq 2\pi$

c. $y = \sin\left(\frac{1}{2}x\right) + 1$ for $-\pi \leq x \leq \pi$

Hint

Take note of the domain given for each graph. When the domain includes values in terms of π, you must sketch the graph in radians.

4. Without using a calculator find the exact values of these trigonometric ratios.

a. $\cos\left(\frac{9\pi}{3}\right)$

b. $\tan\left(-\frac{3\pi}{4}\right)$

c. $\sin\left(-\frac{11\pi}{6}\right)$

Hint

Start by expressing each part in terms of trigonometric ratios of values between 0 and π.

5. E/P Sketch the following graphs on separate sets of axes, marking any points where the graph intersects with the coordinate axes.

a. $y = 2\cos 2x$ for $-2\pi \leq x \leq 0$ (3 marks)

b. $y = \sin\left(x - \frac{\pi}{4}\right)$ for $-\pi \leq x \leq \pi$ (3 marks)

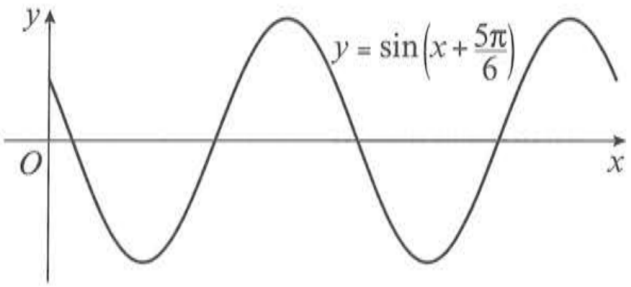
6. E In the triangle PQR, $PQ = 12\,\text{cm}$, $QR = 8\,\text{cm}$ and $RP = 9\,\text{cm}$.

a. Find the size of angle QRP, giving your answer in radians to 3 significant figures. (3 marks)

b. Find the area of triangle PQR, giving your answer in cm^2 to 3 significant figures. (3 marks)

7. E/P The diagram shows the curve with equation

$$y = \sin\left(x + \frac{5\pi}{6}\right), \quad 0 \leq x \leq 4\pi$$

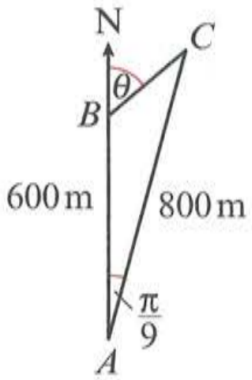


Write down the exact coordinates of the points where the curve intersects with:

a. the x-axis (3 marks)

b. the y-axis. (1 mark)

8. E/P The diagram shows three ships A, B and C. The ships are in the same horizontal plane.
Ship B is 600 m due north of ship A.
Ship C is 800 m from ship A on a bearing of $\frac{\pi}{9}$ radians.



a. Find the distance between ship B and ship C in metres to 3 significant figures. (3 marks)

b. The bearing of ship C from ship B is θ. Find the value of θ in degrees to 3 significant figures. (3 marks)

5.2 Arc length

Practice Book, printed page 36.

1. An arc AB of a circle, with centre O and radius r cm, subtends an angle θ radians at O. Giving exact values where possible, find the length of AB, l cm, when:

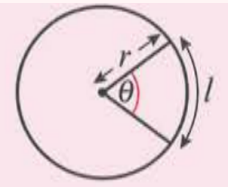
a. $r = 18$, $\theta = 2.5$ rad

b. $r = 6.5$, $\theta = \dfrac{2\pi}{3}$

c. $r = 2.3$, $\theta = 0.35$ rad

Hint

Use $l = r\theta$ to find the arc length.



2. An arc AB of a circle, with centre O and radius r cm, subtends an angle θ radians at O. The length of AB is l cm.
Giving your answers to 3 significant figures, find the radius, r cm, when:

a. $l = 5\sqrt{3}$, $\theta = \dfrac{\pi}{6}$

b. $l = 20$, $\theta = 1.8$ rad

c. $l = 9.25$, $\theta = 0.45$ rad

Hint

Make sure your calculator is in radians mode when using the formula for arc length.

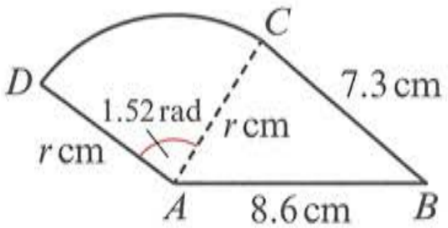
3. An arc AB of a circle, with centre O and radius r cm, subtends an angle θ radians at O. The length of AB is l cm.
Giving your answers to 3 significant figures, find the angle, θ radians, when:

a. $l = 12.1$, $r = 3.4$

b. $l = 7.235$, $r = 0.92$

c. $l = \sqrt{5}$, $r = \sqrt{2}$

4. E/P The shape $ABCD$ consists of a triangle ABC joined to a sector ACD of a circle with radius r cm and centre A. The lines $AB = 8.6$ cm and $BC = 7.3$ cm, and angle $DAC = 1.52$ radians.

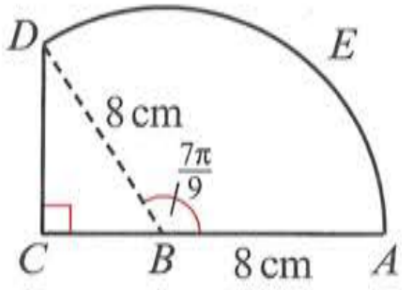


Given that the length of the arc DC is 6.08 cm, find:

a. the length r (2 marks)

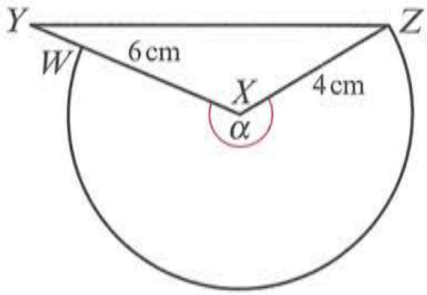
b. the perimeter of the shape $ABCD$. (2 marks)

5. E/P The shape $ABCDEA$ consists of a right-angled triangle BCD joined to a sector $ABDEA$ of a circle with radius 8 cm and centre B.
The points A, B and C lie on a straight line with $AB = 8$ cm and the size of the angle ABD is $\frac{7\pi}{9}$ radians.



a. Calculate the exact length of the arc DEA, giving your answer as a multiple of π. (2 marks)
b. Calculate the perimeter of the shape $ABCDEA$ to 3 significant figures. (4 marks)

6. E/P The shape comprises a triangle XYZ, with $XY = 6$ cm and $ZX = 4$ cm, and part of a circle. The circular arc ZW is a major arc of the circle with centre X and radius 4 cm. The point W lies on the line XY and the angle $ZXW = \alpha$ radians.



Given that the exact length of arc ZW is $\frac{40\pi}{9}$ cm, find:
a. the exact value of α in radians (2 marks)
b. the area of triangle XYZ, giving your answer to 1 decimal place. (3 marks)

7. E/P A circular cake is cut into 12 exactly equal slices, so that the horizontal cross-section of each slice is a sector of a circle.
Given that the arc length of the cross section of each slice is $\frac{5\pi}{3}$ cm, find:
a. the diameter of the cake (2 marks)
b. the exact angle, in radians, between the two straight edges of each slice. (1 mark)

5.3 Areas of sectors and segments

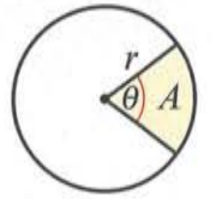
Practice Book, printed page 37.

1.
a. Find the area of the sector of a circle of radius 5 cm, given that the sector subtends an angle of 0.45 radians at the centre of the circle.
b. Find the area of the sector of a circle of radius 7.3 cm, given that the sector subtends an angle of 1.2 radians at the centre of the circle.
c. Find the radius, r cm, of a circle, given that a sector of the circle has an area of $\frac{9\pi}{8}$ cm^2 and the sector subtends an angle of $\frac{\pi}{4}$ radians at the centre of the circle.
d. Find the angle, θ radians, subtended by a sector at the centre of a circle, given that the circle

has radius 7.2 cm and the sector has an area of 33.696 cm^2.

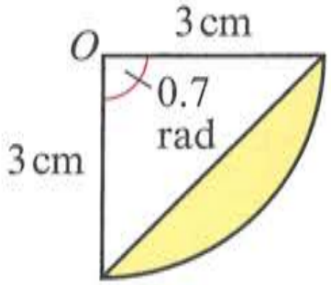
Hint

Draw a diagram of each sector with the given information.
Use $A = \frac{1}{2}r^2\theta$ to find sector area.

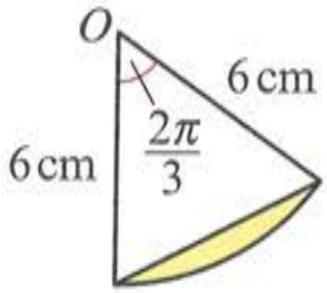


2. Find the area of the shaded segment in each of these sectors.

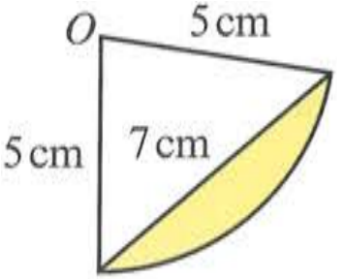
a.



b.



c.

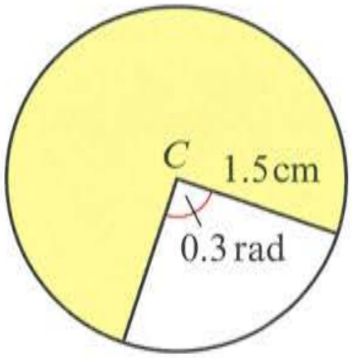


Hint

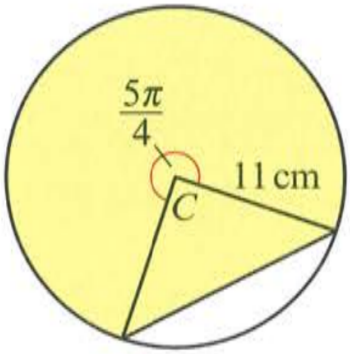
Use the formula $A = \frac{1}{2}r^2(\theta - \sin\theta)$
In part c you need to find θ first. You can do this by using the cosine rule.

3. Find the shaded area in each of these circles with centre C.

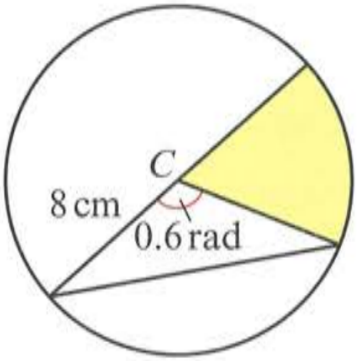
a.



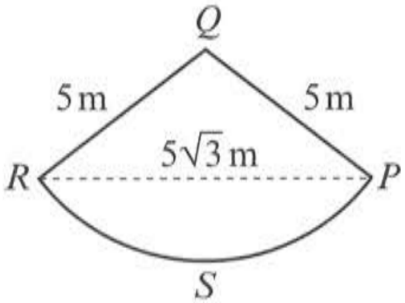
b.



c.



4. E The sector $PQRS$ is part of a circle with centre Q and radius 5 m.
The length of the line segment PR is $5\sqrt{3}$ m.

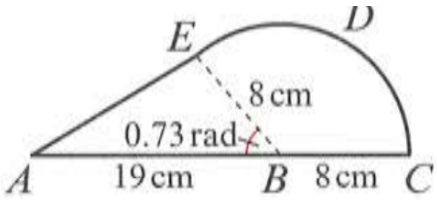


a. Calculate the exact size of angle PQR in radians. (3 marks)

b. Calculate the exact area of the sector, giving your answer as a multiple of π. (3 marks)

c. Calculate the area of the segment PRS, giving your answer to 3 significant figures. (3 marks)

5. E/P The shape consists of a triangle ABE joined to a sector $BCDE$ of a circle with radius 8 cm and centre B. The points A, B and C lie on a straight line with $AB = 19$ cm and $BC = 8$ cm.

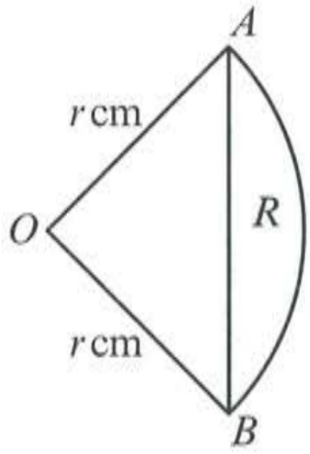


Given that the size of angle ABE is exactly 0.73 radians, find, giving your answers to 3 significant figures:

a. the area of the shape (4 marks)

b. the perimeter of the shape. (5 marks)

6. E/P The diagram shows the sector OAB of a circle of radius r cm.
The area of the sector is $37.8\,\text{cm}^2$ and angle AOB is 1.2 radians.

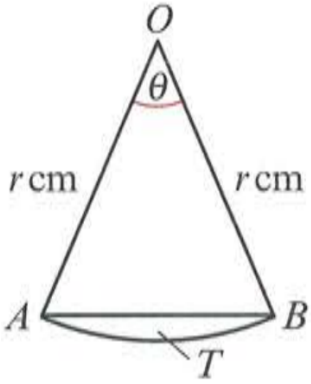


a. Show that $r = 3\sqrt{7}$ cm. (3 marks)

The shaded segment R is enclosed by the arc AB and the straight line AB.

b. Find the area of R, giving your answer to 1 decimal place. (3 marks)

7. E/P The diagram shows a sector AOB of a circle with centre O and radius r cm. The angle AOB is θ radians. The area of the sector AOB is $12\,\text{cm}^2$.



a. Given that the perimeter of the sector is 6 times the length of the arc AB, find the exact values of θ and r. (4 marks)

The shaded segment T is enclosed by the straight line AB and the arc AB.

b. Use your values of θ and r to find the area of T, giving your answer to 3 significant figures. (3 marks)

5.4 Solving trigonometric equations

Practice Book, printed page 39.

1. Solve these equations for θ in the interval $0 \leq \theta \leq 2\pi$, giving your answers to 3 significant figures when they are not exact.

a. $\sin\theta = 0.5$

b. $\tan\theta = -2$

c. $5\cos\theta = 3$

d. $\sqrt{2}\sin\theta + 1 = 2$

Hint

Make sure your calculator is in radians mode.

Hint

For parts c and d, rewrite the equations in the forms $\cos\theta = \ldots$ and $\sin\theta = \ldots$

2. Solve these equations for θ in the given intervals, giving your answers to 3 significant figures when they are not exact.

a. $2 + 3\sin\theta = 4$, $0 \leq \theta \leq \pi$

b. $6\cos\theta - 1 = -4$, $-\pi \leq \theta \leq 2\pi$

c. $\sqrt{2}\tan\theta - 7 = -5$, $-2\pi \leq \theta \leq \pi$

d. $\sqrt{2}\cos 2\theta - 0.6 = 0.4$, $\pi \leq \theta \leq 3\pi$

Hint

Check that your final values are within the given range.

3. Solve these equations for θ in the interval $0 \leq \theta \leq 2\pi$, giving your answers to 3 significant figures.

a. $\tan^2\theta + 3\tan\theta = -2$

b. $3\sin^2\theta - 2\cos\theta + 1 = 0$

c. $\cos\left(\theta + \frac{\pi}{4}\right) = 0.5$

d. $\sin(2\theta + 0.6) = 1$

Hint

For part a, use the substitution $u = \tan\theta$ to form a quadratic equation.
For part b, you need to use the identity $\sin^2\theta + \cos^2\theta \equiv 1$

4. E/P Find all the values of θ in the interval $0 \leq \theta \leq 2\pi$ for which:

a. $\cos\left(\theta - \frac{\pi}{18}\right) = \cos\left(\frac{\pi}{12}\right)$ (3 marks)

b. $\tan 2\theta = 0.6$ (3 marks)

Give your answers to 3 significant figures where they are not exact.

5. E/P

a. Given that $4\sin x = 3\cos x$, find the value of $\tan x$. (1 mark)

b. Solve $4\sin 2x = 3\cos 2x$ for $-\pi \leq x \leq \pi$, giving your answers to 1 decimal place. (5 marks)

6. E/P

a. Show that the equation $2\cos^2 x + 3\sin x = 3$ can be written in the form $2\sin^2 x - 3\sin x + 1 = 0$ (2 marks)

b. Solve $2\sin^2 x - 3\sin x + 1 = 0$ for $0 \leq x \leq \pi$, giving your answers exactly. (4 marks)

7. E/P

a. Solve $\sin\left(3\theta + \frac{\pi}{4}\right) = -0.2$ in the range $0 \leq \theta \leq \frac{3\pi}{2}$
Give your answers to 3 significant figures. (4 marks)

b. Find the exact values of θ in the interval $0 \leq \theta \leq \pi$ for which $4\cos x = 6\tan x$. (6 marks)

5.5 Small angle approximations

Practice Book, printed page 40.

1. When θ is small, find the approximate values of:

a. $\dfrac{\sin 3\theta}{\theta}$

b. $\dfrac{\tan 4\theta}{\sin 2\theta}$

c. $\dfrac{5\tan\theta - \theta}{\sin 4\theta}$

Hint

When θ is small and measured in radians, $\sin\theta \approx \theta$ and $\tan\theta \approx \theta$.

2. When θ is small, find the approximate values of:

a. $\dfrac{1-\cos 2\theta}{\theta\sin\theta}$

b. $\dfrac{\cos 6\theta - 1}{\sin 3\theta\tan\theta}$

c. $\dfrac{\theta\tan\theta}{1-\cos\theta}$

Hint

When θ is small and measured in radians, $\cos\theta \approx 1 - \dfrac{\theta^2}{2}$

3. When θ is small, show that:

a. $\dfrac{2\theta - \sin 3\theta}{\theta\tan 4\theta} = -\dfrac{1}{4}\theta$

b. $\dfrac{\cos\theta - 1}{\sin\theta} = -\dfrac{1}{2}\theta$

c. $\dfrac{\sin\theta + \tan\theta}{1-\cos 2\theta} = \dfrac{1}{\theta}$

Hint

Use the small angle approximations and then simplify.

4. E Given that θ is small and is measured in radians, use the small angle approximations to find an approximate value of

$$\frac{1-\cos 2\theta}{3\theta\sin 2\theta}$$

(3 marks)

5. E/P Given that θ is small, use the small angle approximations for $\sin\theta$, $\cos\theta$ and $\tan\theta$ to show that

$$\frac{3\tan\theta - 4\cos\theta + 5}{\sin\theta + 1} \approx 2\theta + 1$$

(3 marks)

6. E/P

a. Given that θ is small, use the small angle approximation for $\cos\theta$ to show that $5\cos^2\theta + 6\cos\theta + 1 \approx 12 - 8\theta^2$ (3 marks)

Josh tests the approximation in part a on his calculator by using $\theta = 10°$, and his working is shown below.

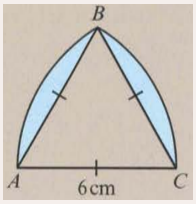
> $5\cos^2 10° + 6\cos 10° + 1 = 11.76$, to 4 significant figures.
> The approximation gives $12 - 8(10)^2 = -788$
> Therefore, $5\cos^2\theta + 6\cos\theta + 1 \approx 12 - 8\theta^2$ is not true for $\theta = 10°$.

b. Is Josh correct? Give full reasons for your answer. (2 marks)

Problem solving: Set A

Bronze

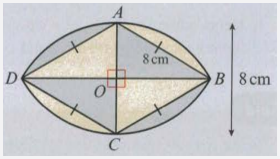
The diagram shows an equilateral triangle ABC with side length 6 cm.
A segment of a circle with centre C is attached to AB, and an identical segment of a circle with centre A is attached to BC.



a. Write down the exact size of angle CAB in radians. (1 mark)
b. Find the exact area of triangle ABC. (2 marks)
c. Find the exact area of the sector with centre A and arc BC, giving your answer as a multiple of π. (2 marks)
d. Hence show that the shaded area of the shape is $(12\pi - 18\sqrt{3})\,\text{cm}^2$. (2 marks)

Silver

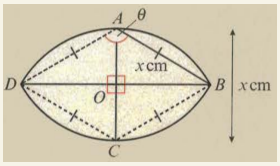
A shape is comprised of a rhombus enclosed by two arcs of circles. Arc BAD has centre C and arc BCD has centre A. Some of the dimensions of the shape are shown in the diagram.



a. Find the exact area of the rhombus $ABCD$. (3 marks)
b. Find the exact area of the sector with centre A and arc BCD, giving your answer as a multiple of π. (3 marks)
c. Hence show that the shaded area of the shape is $\frac{1}{3}(128\pi - 96\sqrt{3})\,\text{cm}^2$. (2 marks)

Gold

A shape is comprised of a rhombus surrounded by two arcs. Arc BAD has centre C and arc BCD has centre A. Angle DAB is θ radians.



Show that the area of the shape is
$\left(\theta - \frac{\sqrt{a}}{b}\right)x^2$
where a and b are integers to be found. (8 marks)

Problem solving: Set B

Bronze

a. Sketch the following graphs on the same set of axes:
i. $y = \tan x$ for $0 \leq x \leq 2\pi$
ii. $y = \frac{1}{3}$
iii. $y = -3$
(3 marks)

b. Show that the equation $3\tan^2 x + 8\tan x - 3 = 0$ can be written as $(3\tan x - 1)(\tan x + 3) = 0$ (1 mark)

c. Hence solve the equation $3\tan^2 x + 8\tan x - 3 = 0$ for $0 \leq x \leq 2\pi$, giving your answers to 3 significant figures. (3 marks)

Silver

a. Show that $3\cos^2 x = 5(1 - \sin x)$ can be written as $3\sin^2 x - 5\sin x + 2 = 0$ (3 marks)

b. Hence solve $3\cos^2 x = 5(1 - \sin x)$ for $-\pi \leq x \leq \pi$, giving your answers to 3 significant figures where they are not exact. (4 marks)

Gold

Solve $\cos x = 3\tan x$ for $0 \leq x \leq 3\pi$, giving your answers to 3 significant figures. (8 marks)

Now try this ⟶ Exam question bank Q25, Q30, Q35, Q41, Q44, Q46

Selected exam question bank questions

Question 25

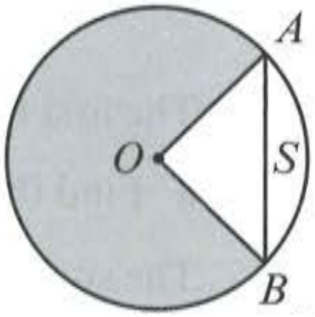
a. When θ is small, show that $\dfrac{1+\sin 2\theta+\tan 2\theta}{2\cos 4\theta-1} \approx \dfrac{1+4\theta}{(1+2\theta)(1-2\theta)}$ (4 marks)

b. Hence write down an estimate of $\dfrac{1+\sin 2\theta+\tan 2\theta}{2\cos 4\theta-1}$ when $\theta = 0.03$ radians. (1 mark)

c. Find the percentage error in your estimate, giving your answer to 3 significant figures. (2 marks)

Question 30

The diagram shows a circle with centre O. The points A and B lie on the circumference of the circle. The area of the major sector, shown shaded, is $88.725\,\text{cm}^2$. The reflex angle AOB is 4.2 radians.



a. Find the exact length, in cm, of the minor arc AB, giving your answer in the form $a\pi + b$, where a and b are constants to be found. (4 marks)

b. Find the area of the segment S, giving your answer to 1 decimal place. (3 marks)

Question 35

Luke and Isabella were given this question.

Solve the equation $\sqrt{3}\sin\theta = \cos\theta$ for $-\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2}$

Their attempts are shown below.

Isabella

$\sqrt{3}\sin\theta = \cos\theta$

$\tan\theta = \sqrt{3}$

$\theta = \dfrac{\pi}{3}$

Luke

$\sqrt{3}\sin\theta = \cos\theta$

$3\sin^2\theta = \cos^2\theta$

$3\sin^2\theta = 1 - \sin^2\theta$

$\sin^2\theta = \dfrac{1}{4}$

$\sin\theta = \pm\dfrac{1}{2}$

$\theta = \pm\dfrac{\pi}{6}$

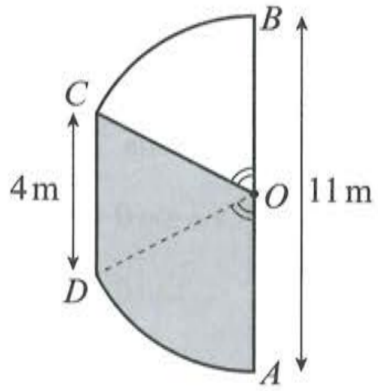
a. What error has Isabella made? (1 mark)

Luke gives $\theta = -\dfrac{\pi}{6}$ as one of the answers to $\sqrt{3}\sin\theta = \cos\theta$

b. Explain why this answer is incorrect and how this answer arose. (2 marks)

Question 41

The shape $AOBCD$ consists of an isosceles triangle COD joined along its equal sides to sectors OBC and ODA of a circle with centre O and diameter 11 m. Angles AOD and BOC are equal. AOB is a straight line and is parallel to the line DC. DC has length 4 m.



a. Find angle COD in radians, correct to 3 decimal places. (2 marks)

b. Find the shaded area $AOCD$, giving your answer to 1 decimal place. (3 marks)

c. Find the perimeter of sector COB, giving your answer to 1 decimal place. (2 marks)

Question 44

Consider the function $\mathrm{f}(x) = -2x^3 + 8x^2 - 7x + 3$, $x \in \mathbb{R}$.

a. Given that $\mathrm{f}(3) = 0$, write $\mathrm{f}(x)$ as a product of two algebraic factors. (2 marks)

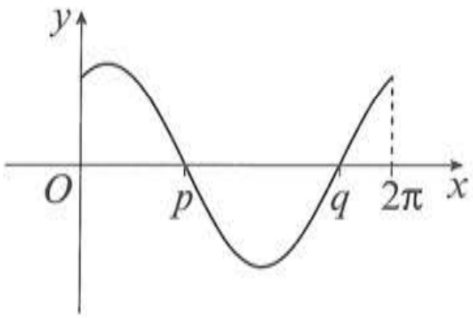
b. Hence prove that there are exactly two real solutions to the equation $-2y^6 + 8y^4 - 7y^2 + 3 = 0$ (2 marks)

c. Deduce the number of real solutions, for $5\pi \leq \theta < 8\pi$, to the equation $2\tan^3\theta - 8\tan^2\theta + 7\tan\theta - 3 = 0$ (1 mark)

Question 46

The diagram shows the curve with equation $y = k\sin\left(x + \frac{\pi}{3}\right)$ for $0 \leq x \leq 2\pi$, where k is a constant.

The curve cuts the y-axis at $(0, 2\sqrt{3})$ and passes through the points $(p, 0)$ and $(q, 0)$.



a. Show that $k = 4$. (1 mark)

b. Find p and q. (2 marks)

The line $y = -1.2$ meets the curve at the points R and S.

c. Find the x-coordinates of R and S, giving your answers to 3 significant figures. (5 marks)

Answers

5.1 Radian measure

1.

a. $150°$

b. $22.5°$

c. $720°$

d. $19.5°$

e. $81.0°$

f. $108.9°$

2.

a.

i. $\frac{\pi}{10}$

ii. $\frac{13\pi}{72}$

iii. $\frac{16\pi}{9}$

b.

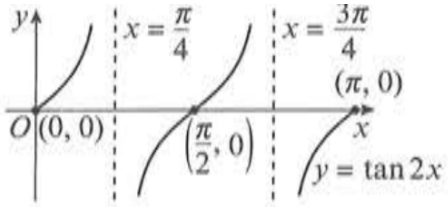
i. 0.279 rad

ii. 2.16 rad

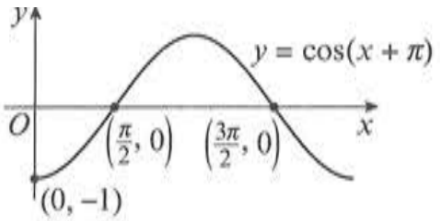
iii. 4.54 rad

3.

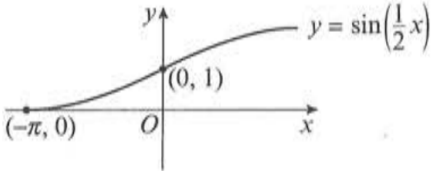
a.



b.



c.



4.

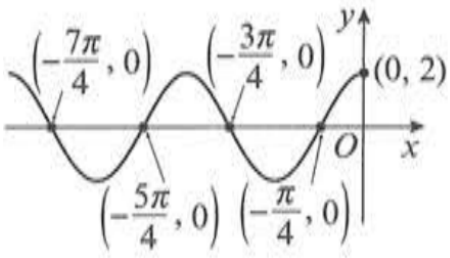
a. -1

b. 1

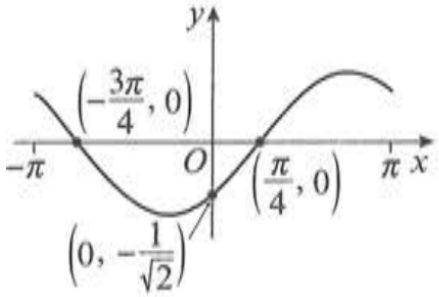
c. $\frac{1}{2}$

5.

a.



b.



6.

a. 1.56 rad

b. $36.0\,\text{cm}^2$

7.

a. $\left(\frac{\pi}{6}, 0\right)$, $\left(\frac{7\pi}{6}, 0\right)$, $\left(\frac{13\pi}{6}, 0\right)$, $\left(\frac{19\pi}{6}, 0\right)$

b. $\left(0, \frac{1}{2}\right)$

8.

a. 313 m

b. 061.0°

5.2 Arc length

1.

a. 45 cm

b. $\frac{13\pi}{3}$ cm

c. 0.805 cm

2.

a. 16.5 cm

b. 11.1 cm

c. 20.6 cm

3.

a. 3.56 rad

b. 7.86 rad

c. 1.58 rad

4.

a. 4 cm

b. 25.98 cm

5.

a. $\frac{56\pi}{9}$ cm

b. $38.8\,\text{cm}$

6.

a. $\dfrac{10\pi}{9}$

b. $4.1\,\text{cm}^2$

7.

a. $20\,\text{cm}$

b. $\dfrac{\pi}{6}$

5.3 Areas of sectors and segments

1.

a. $5.63\,\text{cm}^2$ (3 s.f.)

b. $32.0\,\text{cm}^2$ (3 s.f.)

c. $3\,\text{cm}$

d. 1.3 rad

2.

a. $0.251\,\text{cm}^2$ (3 s.f.)

b. $22.1\,\text{cm}^2$ (3 s.f.)

c. $6.89\,\text{cm}^2$ (3 s.f.)

3.

a. $6.73\,\text{cm}^2$ (3 s.f.)

b. $280.0\,\text{cm}^2$ (3 s.f.)

c. $81.3\,\text{cm}^2$ (3 s.f.)

4.

a. $\dfrac{2\pi}{3}$

b. $\dfrac{25\pi}{3}\,\text{m}^2$

c. $15.4\,\text{m}^2$

5.

a. $128\,\text{cm}^2$

b. $60.4\,\text{cm}$

6.

a. $r = \sqrt{\dfrac{2 \times 37.8}{1.2}} = 3\sqrt{7}\,\text{cm}$

b. $8.4\,\text{cm}^2$

7.

a. $\theta = \dfrac{2}{5}$, $r = 2\sqrt{15}\,\text{cm}$

b. $0.317\,\text{cm}^2$

5.4 Solving trigonometric equations

1.

a. $\theta = \dfrac{\pi}{6}, \dfrac{5\pi}{6}$

b. $\theta = 2.03, 5.18$

c. $\theta = 0.927, 5.36$

d. $\theta = \dfrac{\pi}{4}, \dfrac{3\pi}{4}$

2.

a. $\theta = 0.730, 2.41$

b. $\theta = -\frac{2\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}$

c. $\theta = -5.33, -2.19, 0.955$

d. $\theta = \frac{9\pi}{8}, \frac{15\pi}{8}, \frac{17\pi}{8}, \frac{23\pi}{8}$

3.

a. $\theta = 2.03, 2.36, 5.18, 5.50$

b. $\theta = 0.519, 5.76$

c. $\theta = \frac{\pi}{12}, \frac{17\pi}{12}$

d. $\theta = 0.485, 3.63$

4.

a. $\theta = \frac{5\pi}{36}, \frac{71\pi}{36}$

b. $\theta = 0.270, 1.84, 3.41, 4.98$

5.

a. $\tan x = \frac{3}{4}$

b. $x = -2.8, -1.2, 0.3, 1.9$

6.

a. $2(1 - \sin^2 x) + 3\sin x - 3 = 0 \Rightarrow 2\sin^2 x - 3\sin x + 1 = 0$

b. $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$

7.

a. $\theta = 0.853, 1.77, 2.95, 3.86$

b. $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$

5.5 Small angle approximations

1.

a. 3

b. 2

c. 1

2.

a. 2

b. -6

c. 2

3.

a. $\frac{-\theta}{4\theta^2} = -\frac{1}{4\theta}$

b. $\frac{\left(1 - \frac{\theta^2}{2}\right) - 1}{\theta} = \frac{-\frac{\theta^2}{2}}{\theta} = -\frac{1}{2}\theta$

c. $\frac{2\theta}{1 - \left(1 - \frac{(2\theta)^2}{2}\right)} = \frac{2\theta}{2\theta^2} = \frac{1}{\theta}$

4. $\frac{1}{3}$

5. $\frac{3\theta - 4\left(1 - \frac{\theta^2}{2}\right) + 5}{\theta + 1} = \frac{3\theta - 4 + 2\theta^2 + 5}{\theta + 1}$

$= \dfrac{(2\theta+1)(\theta+1)}{\theta+1} = 2\theta+1$

6.

a. $1+6\cos\theta+5\cos^2\theta \approx 1+6\left(1-\dfrac{1}{2}\theta^2\right)+5\left(1-\dfrac{1}{2}\theta^2\right)^2$

$= 1+6\left(1-\dfrac{1}{2}\theta^2\right)+5\left(1-\theta^2+\dfrac{1}{4}\theta^4\right)$

$= 1+6-3\theta^2+5-5\theta^2+\dfrac{5}{4}\theta^4$

$= 12-8\theta^2$

b. Josh is incorrect as he used degrees and not radians. He should substitute $\theta = \dfrac{10\pi}{180}$ and not $\theta = 10$:

$12-8\left(\dfrac{10\pi}{180}\right)^2 = 11.76$

So $\theta = 10^\circ$ gives a good approximation.

Problem solving: Set A

Bronze

a. $\dfrac{\pi}{3}$

b. $9\sqrt{3}\,\text{cm}^2$

c. $6\pi\,\text{cm}^2$

d. $2(6\pi-9\sqrt{3}) = (12\pi-18\sqrt{3})\,\text{cm}^2$

Silver

a. $32\sqrt{3}\,\text{cm}^2$

b. $\dfrac{64\pi}{3}\,\text{cm}^2$

c. Area of shaded part of shape

$=$ area of sector $-\dfrac{1}{2}\times$ area of rhombus

$= \left(\dfrac{64\pi}{3}-16\sqrt{3}\right)\text{cm}^2 = \dfrac{1}{3}(64\pi-48\sqrt{3})\text{cm}^2$

Gold

Area of the sector $ADCB = \dfrac{1}{2}\times x^2\times\theta = \dfrac{\theta}{2}x^2$

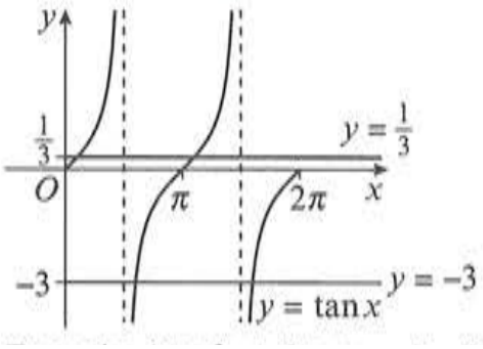
Area of rhombus $= 2\left(\dfrac{1}{2}\times\sqrt{3}x\times\dfrac{x}{2}\right) = \dfrac{\sqrt{3}}{2}x^2$

Total area $= \dfrac{\theta}{2}x^2+\left(\dfrac{\theta}{2}x^2-\dfrac{\sqrt{3}}{2}x^2\right) = \left(\theta-\dfrac{\sqrt{3}}{2}\right)x^2$

Problem solving: Set B

Bronze

a.



b. Factorise $3\tan^2 x + 8\tan x - 3 = 0$ to get
$(3\tan x - 1)(\tan x + 3) = 0$

c. $0.322, 1.89, 3.46, 5.03$

Silver

a. $3\cos^2 x = 5(1 - \sin x) \Rightarrow 3(1 - \sin^2 x) = 5(1 - \sin x)$
$\Rightarrow 3 - 3\sin^2 x = 5 - 5\sin x \Rightarrow 3\sin^2 x - 5\sin x + 2 = 0$

b. $x = 0.730, 2.41$ and $\dfrac{\pi}{2}$

Gold $x = 0.308, 2.83, 6.59, 9.11$

Selected exam question bank answers

Question 25

a. $\dfrac{1 + \sin 2\theta + \tan 2\theta}{2\cos 2\theta - 1} = \dfrac{1 + 4\theta}{2\left(1 - \frac{(2\theta)^2}{2}\right) - 1}$
$= \dfrac{1 + 4\theta}{(1 - 4\theta)^2} = \dfrac{1 + 4\theta}{(1 + 2\theta)(1 - 2\theta)}$

b. 1.124 (3 d.p.)

c. $3.12 \times 10^{-3}\%$

Question 30

a. $(13\pi - 27.3)$ cm

b. $25.6\,\text{cm}^2$

Question 35

a. She has used $\dfrac{\cos\theta}{\sin\theta} = \tan\theta$
She should have used $\dfrac{\sin\theta}{\cos\theta} = \tan\theta$

b. $\sqrt{3}\sin\left(-\dfrac{\pi}{6}\right) \neq \cos\left(-\dfrac{\pi}{6}\right)$, so can't be a solution.
Error was caused by squaring.

Question 41

a. 0.744

b. $28.4\,\mathrm{m}^2$

c. $17.6\,\mathrm{m}$

Question 44

a. $(x-3)(-2x^2+2x-1)$

b. $-2y^6+8y^4-7y^2+3=(y^2-3)(-2y^4+2y^2-1)$
$-2y^4+2y^2-1=0$ has no real solutions
$y^2-3=0$ has 2 real solutions: $y=\pm\sqrt{3}$

c. 3

Question 46

a. $2\sqrt{3}=k\sin\left(0+\dfrac{\pi}{3}\right)\Rightarrow k=4$

b. $p=\dfrac{2\pi}{3}, q=\dfrac{5\pi}{3}$

c. $2.40, 4.93$